

3.4.2 P-series Worksheet

Determine whether each of the following series converges or diverges. You may use any theorem from 3.4.2 or earlier.

1.
$$\sum_{n=1}^{\infty} \left(\frac{2}{n^2} - \frac{2}{2^n} \right)$$

2.
$$\sum_{n=1}^{\infty} \frac{4^{n-1}}{2^n}$$

3.
$$\sum_{n=1}^{\infty} \frac{\sqrt{n} + 4}{n^2}$$

$$4. \sum_{n=1}^{\infty} \frac{1}{n^2 + 4}.$$

$$5. 1 + \frac{2}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} + \frac{1}{4\sqrt{4}} + \frac{1}{5\sqrt{5}} + \dots$$

$$6. \sum_{k=1}^{\infty} \frac{7}{3k}$$

$$7. \sum_{i=1}^{\infty} \operatorname{sech}^2(i)$$

$$8. \sum_{k=1}^{\infty} \frac{4k^3 + 2k}{2k^3 - 1}$$

9. $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2}$